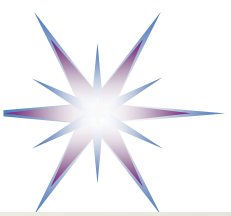


GenAI/LLM as a Researcher Tool

Assisting in The Structured Systematic Literature Review

Methodology development and grounding for System and Software Engineering

Yuri Demchenko
IvI, UvA
NTUU “Igor Sikorsky KPI”



Outlook

- Knowledge complexity: Presentation, Representation, Perception
- Knowledge Complexity and Epistemic Governance Model of LLM-Assisted Research (KCEG)
- Literature review methodologies
- Systematic Literature Review (SLR)
 - Workflow and reporting
 - Structured SLR – Extending Left and Right
- SLR01 and SLR02 case studies
- Additional slides
 - Mapping SLR process to KCEG and HGLEP

Disclaimer

This is a working material to advise both PhD/doctoral students and provide information for advisers on best practices how to do literature reviews, reflecting current data management and GenAI technologies.

Recommended reading paths:

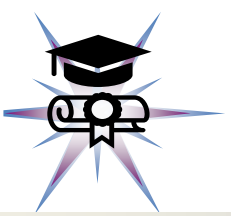
For advisers – the whole material

For doctoral student – slides marked with icon ->



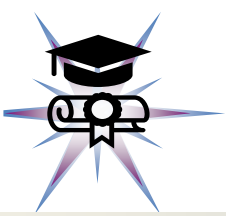
Warning for doctoral students

The proposed methodology is aided for primarily being used as a manual researcher cognitive activity at the initial stages up to Stage 3, starting from which you can and will be able effectively use GenAI/LLM with benefit for building your research body of knowledge



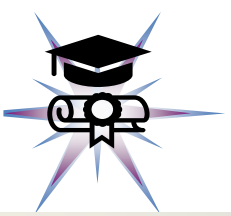
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 - Thematic story, expert interpretation
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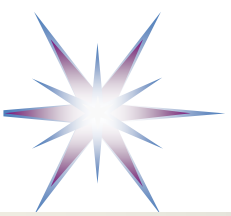
Existing Recommendations for Systematic Review (+ Scoping Reviews and Mapping Study)

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- [2] Kitchenham, B., Mendes, E., Travassos, G.H. (2007) A Systematic Review of Cross- vs. Within-Company Cost Estimation Studies, IEEE Trans on SE, 33 (5) https://www.researchgate.net/publication/3189730_Cross_versus_Within-Company_Cost_Estimation_Studies_A_Systematic_Review
- [3] Claes Wohlin , Guidelines for **Snowballing** in Systematic Literature Studies and a Replication in Software Engineering, Proc. EASE '14, May 13 - 14 2014, London, England, BC, United Kingdom, <https://www.wohlin.eu/ease14.pdf>
- [4] Pai, M., McCulloch, M, Gorman, J.D., Pai, N., Enanoria, W., Kennedy, G., Tharyan, P. and Colford, J.M. Jr. (2004) Systematic reviews and meta-analyses: An illustrated step-by-step guide. *The National Medical Journal of India*, **17**(2), 86-95.
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- [7] Sai Kaushik Upputuru, Cameron John Sabet, Trisha Bhathivi, Dang Nguyen, Exploring Scoping and Mapping Reviews, In book: Practical Approach to Conducting a Systematic Review and Meta-Analysis, DOI: 10.4018/979-8-3693-3007-4.ch002, July 2025



Knowledge Complexity Aspects

- Knowledge complexity views: presentation, representation and perception
- General Knowledge Complexity aspects linked to knowledge presentation
 - Vocabulary and semantics describe **how knowledge is expressed** (in words and terms)
 - Ontology and structure describe **how it is organised**
 - Inference and methodology describe **how it works and how it is produced**
 - Epistemic measures describe **how reliable, novel, uncertain, or contested it is**
- For humans, complexity is strongly tied to prior knowledge, cognitive load, and research judgment, in addition to tacit knowledge.
- For LLMs, complexity is tied to representation, grounding, context, retrieval, reasoning depth, and verifiability.
- The measure cannot be a single score but a **complexity vector** across identified dimensions.



Levels of complexity in knowledge representation and link to knowledge perception

Knowledge representation/production

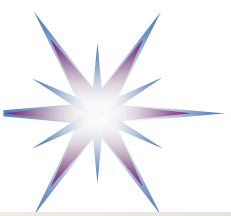
1. Vocabulary / terminology (K1)
2. Syntax / expression form (K2)
3. Factual / propositional layer (K3)
4. Semantic layer (K4)
5. Ontological layer (K5)
6. Structural / graph layer (K6)
7. Inferential layer (K7)
8. Methodological layer (K8)
9. Epistemic / research-frontier layer (K9)

Knowledge complexity = lexical complexity + semantic complexity + ontological complexity + structural complexity + inferential complexity + methodological complexity + epistemic uncertainty

Knowledge complexity for/with humans (perception – general cognitive load)

- **Conceptual load:** how many concepts must be held together.
- **Abstraction level:** concrete observations → variables → models → theories → meta-theories.
- **Methodological load:** how many experimental, statistical, computational, or interpretive steps are needed.
- **Epistemic load:** how much uncertainty, disagreement, missing evidence, or assumption-dependence exists.
- **Creative load:** whether the task is to recall, understand, apply, analyse, evaluate, or create; this resembles the progression in Bloom-style educational taxonomies.

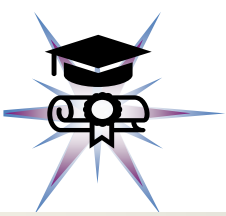
For humans, the hardest knowledge is where the researcher must decide **what counts as evidence, which abstraction is appropriate, which assumptions are defensible, and how to generate a new question.**



Framework:

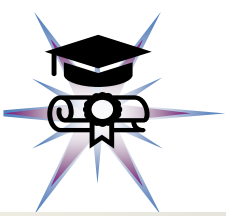
Human-Governed LLM Epistemic Projection for Frontier Research

- **Knowledge-Complexity and Epistemic-Governance Model of LLM-Assisted Research (KCEG)**
- **Human-Governed LLM Epistemic Projection for Frontier Research (HGLEP – HGLEP-FR)**
 - **LLM-assisted research** is most valuable when the LLM acts as an **epistemic projection instrument**: it transforms uneven, large-scale, and partially reliable accessible knowledge into structured **research candidates**, while the expert researcher governs interpretation, verification, and integration into validated knowledge.
 - The researcher remains the **epistemic governor** of the process.
- LLM should not be treated as autonomous producers of validated research knowledge.
 - In frontier research, LLM can **project, reorganize, and structure LLM-accessible knowledge** around a researcher's question.
 - The resulting outputs are not final truths, but **candidate knowledge structures** that require expert interpretation, methodological verification, and epistemic validation.



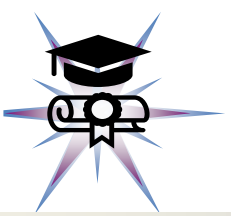
Knowledge Complexity Model as a basis for *Structured* Systematic Literature Review (SSLR)

- Classical Systematic Literature Review (SLR) methodology
- Peer-reviewed papers and other sources of research domain related information
- Missing elements and aspects for Computer Systems and Systems and Software Engineering
- Example and exercise with the detailed paper review
- Structured SLR aligned with KCEG knowledge complexity model



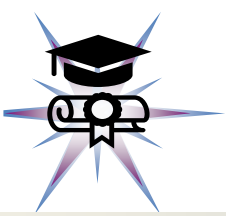
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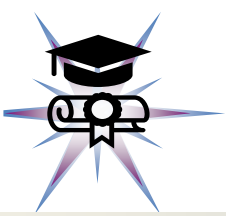
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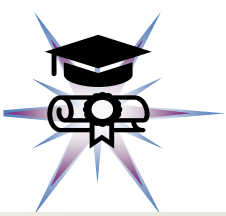
Literature Review (SLR) methodology and specifics for System and Software Engineering (SSE)

- A **Systematic Literature Review** is well a **planned, auditable, reproducible method** for identifying, selecting, evaluating, extracting, and synthesising evidence relevant to a defined research question.
 - In Software Engineering, the classic Kitchenham & Charters guidance explicitly defines SLRs as a way to evaluate and interpret all available research relevant to a research question using a “trustworthy, rigorous, and auditable methodology,” and organises the work into three phases:
 - **planning, conducting, and reporting.**
- **Software Engineering has a well-developed SLR tradition** because of evidence-based software engineering, Kitchenham-style guidelines, systematic mapping studies, and Wohlin-style snowballing practice.
- AI, networking, systems, cybersecurity, HCI, and data management communities often use SLR-like methods, but may call them “surveys,” “systematic mappings,” “taxonomic reviews,” or “structured literature reviews.”
- For a doctoral student in CS/SSE, the safest publishable approach is to combine:
 - **Kitchenham-style SLR protocol + PRISMA-style reporting + CS/SSE database coverage + snowballing + explicit taxonomy/evidence synthesis + transparent GenAI assistance** (at the stage of final report and validation)



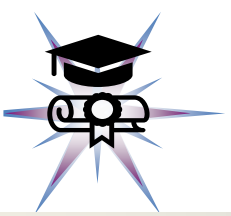
Literature: Peer-reviewed papers and other sources of research domain related information

- Peer reviewed papers is a recommended type of and source of information for initial stage of the doctoral study
 - Recommended sources: IEEE, GoogleScholar, other
 - Self-archiving services: Zenodo, Archiv
- Online technology related publications
- Standards and industry best practices
- Recent regulations (governmental and industry) are becoming an important source of information that shapes technology requirements and compliance
- Expert blogs and self-publishing sites like Medium, also LinkedIn, bring novel information for frontier technologies, like AI, Agentic AI, and Sustainability



Not to miss elements and aspects for Computer Systems and Systems and Software Engineering

- SLR study requires sufficient structured knowledge of the research domain to be achieved in the preparation or iteratively achieved during preparation period
 - SLR is not an educational act but an application of the initial structured knowledge (as pre-requisite for consistent SLR) to identify defensible/contributing research goals
 - For initial domain orientation, other study types to be used (1) narrative or landscape analysis and (2) Scoping review / mapping study
- For SSE, SLR prep/setup beneficially should include the structure (or at least components) of the system or SW applications
 - System & App components/functional blocks + signals input/output + expected functions/operation
 - Beneficially supported with high level diagram/architecture of the system
 - Why? – Good papers contain a graphical presentation of different components of the research (besides testing results).
 - Knowledge of structured diagrams “language” and “semiotics” is beneficial
- All above is required to develop actionable **assessment criteria** and extraction of information sufficient for making taxonomy, classification and summarised assessment, also identify paper usability for research
 - Assessment criteria should be organised in groups
 - Consider revising and updating assessment criteria after initial papers review (close analysis)
 - Why? For easy annotation, ranking and visualisation, also to be **reader-friendly** and **reviewer-aware**.



Typical workflow SLR – General [1] [2]

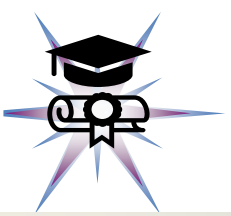
Phase 1 – Planning

Phase 2 – Conducting

- Run database searches
 - Common sources in Computer Science and Software Engineering include ACM Digital Library (central in CS reviews), IEEE Xplore, Scopus, Web of Science, ScienceDirect, SpringerLink, DBLP, and Google Scholar for supplementary checking
- Export and deduplicate records
 - Usually using Zotero, Mendeley, EndNote, Rayyan, Covidence, ASReview, or spreadsheets.
- Title/abstract screening
- Full-text screening
- Read selected papers in detail and record reasons for exclusion
- Backward and forward snowballing - Provenance analysis
- Quality assessment of papers before full full text analysis
- Data extraction (valuable information in the paper, considered important for report synthesis using in own study) – consider/select adding comments to paper in electronic or paper&ink form
- Synthesis
 - This may be quantitative, qualitative, thematic, taxonomic, or mixed. In CS/SSE, meta-analysis is less common than in medicine because studies are often heterogeneous and have a clear evidence (code, experiment data, diagrams)

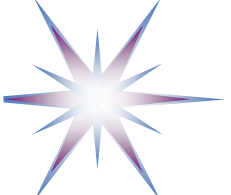
Phase 3 – Reporting the review

- Recommended format **PRISMA 2020** framework - evidence-based reporting guideline designed to ensure transparent, complete, and accurate reporting of systematic reviews and meta-analyses
- 27 points checklist <https://www.prisma-statement.org/prisma-2020-checklist>
- PRISMA Flow diagram <https://www.prisma-statement.org/prisma-2020-flow-diagram>

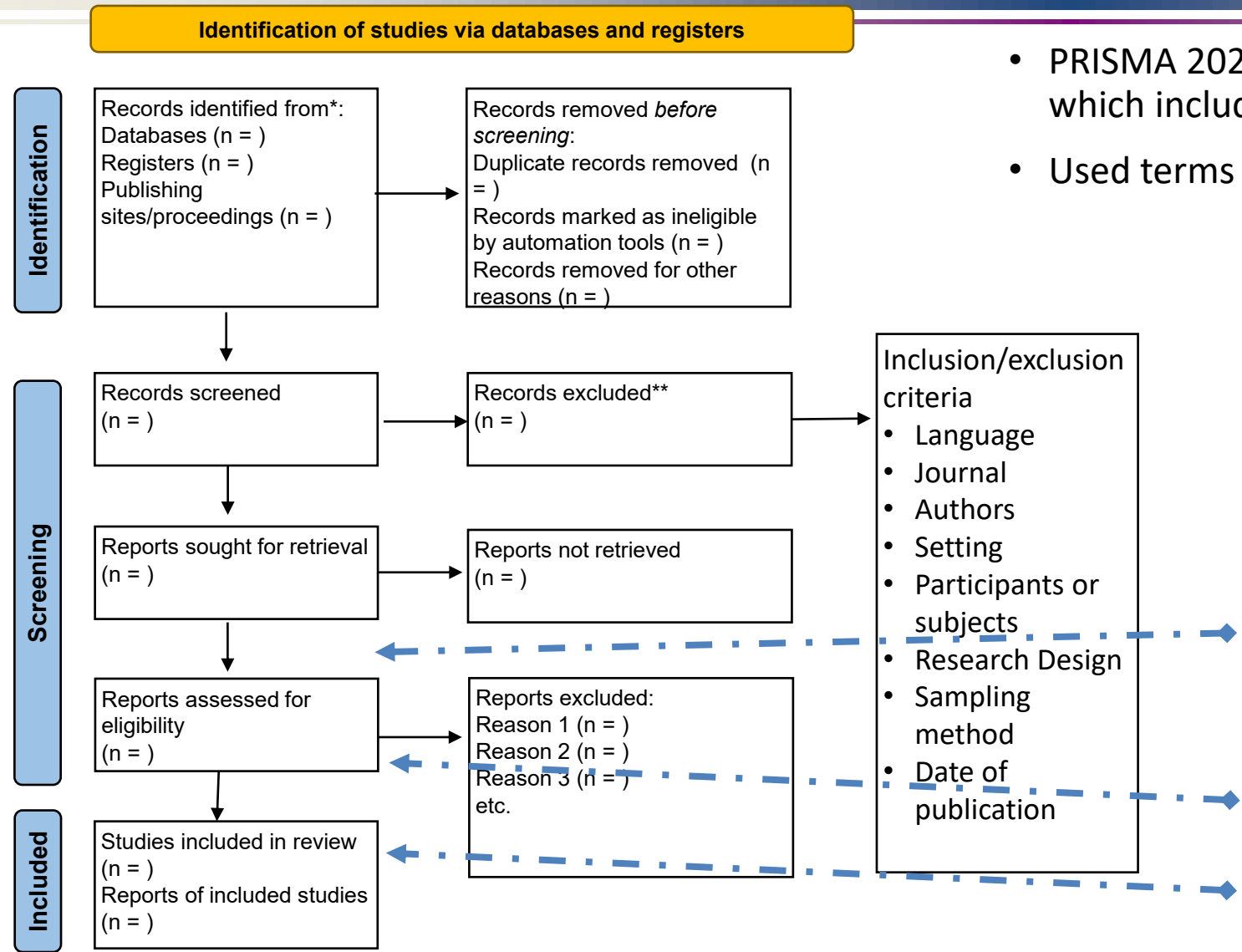


Systematic Literature Review and Provenance - Snowballing

- Backwards and forward snowballing
 - Backward snowballing means checking references
 - Forward snowballing means checking later papers that cite the selected paper.
- Wohlin's Software Engineering guidance treats snowballing as a systematic search method
 - Argues that it can be a strong alternative or complement to database search, especially in SE where terminology and indexing are inconsistent.
 - Helps also to build consistent terminology for new emerging domains or technology re-grounding (migrating to a new technology platform)

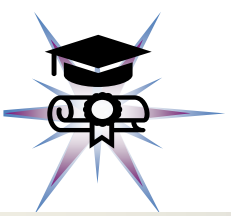


PRISMA 2020 flow diagram for new systematic review reporting



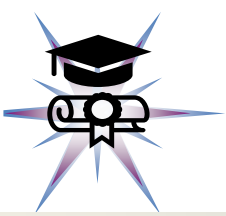
- PRISMA 2020 flow diagram for new systematic reviews, which included searches of databases and registers only
- Used terms clarification

SLR step	PRISMA object	What the doctoral student handles
Database search	Records	Search results from ACM, IEEE, Scopus, WoS, etc.
Deduplication	Records	Remove duplicate database entries
Title/abstract screening	Records	Decide which records may be relevant
Full-text retrieval	Reports	Get the actual papers/documents
Full-text eligibility	Reports	Decide whether the paper/document fits criteria
Data extraction	Studies/ contribution	Extract methods, evidence, tools, models, findings
Synthesis	Studies	Compare and integrate evidence
Final inclusion	Studie/reports	Report both included papers and included evidence units

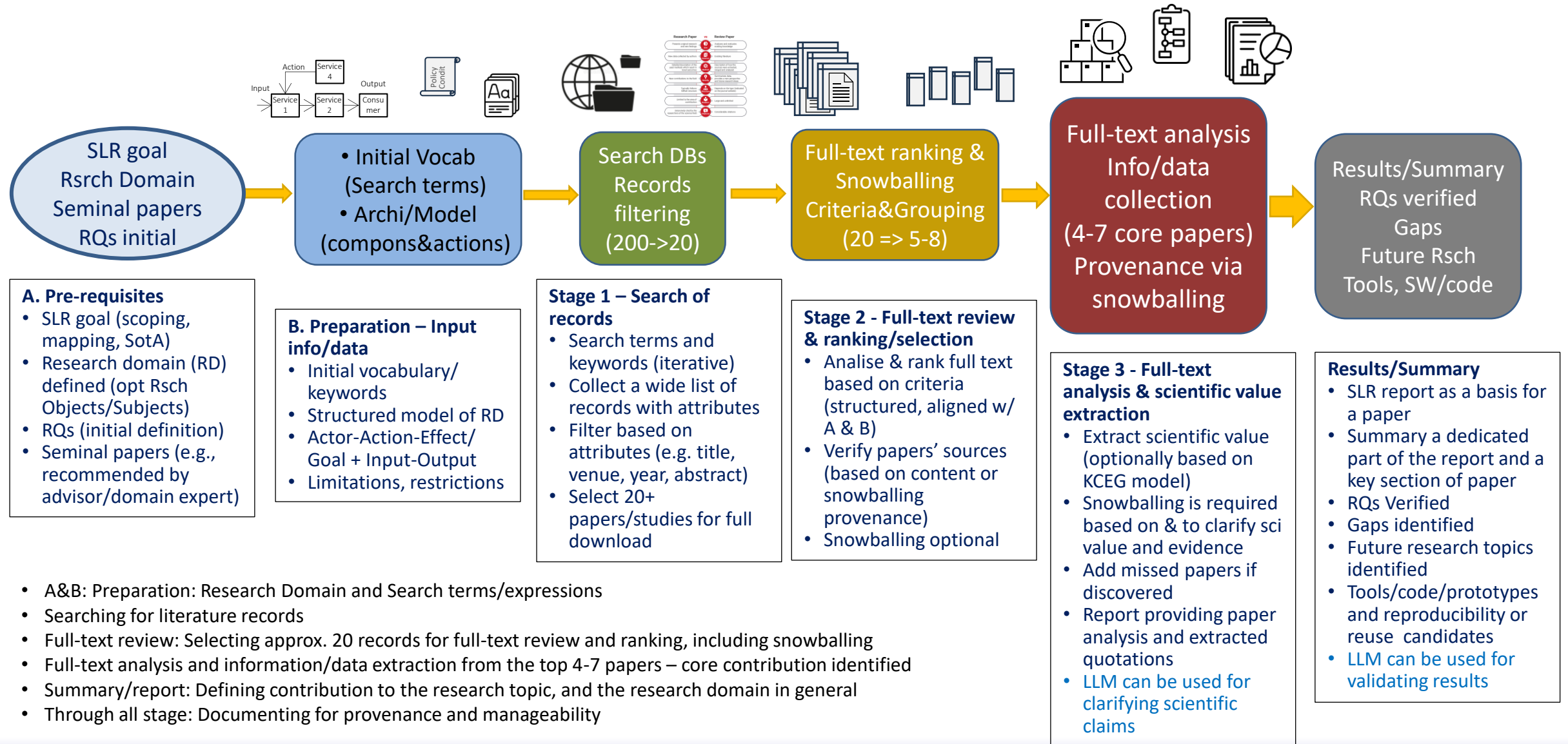


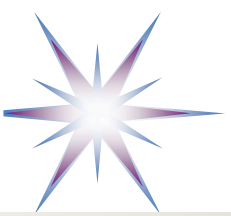
Typical Steps in SLR [ref 2] – Focused and simplified

- Identification of the need for carrying out a systematic review
 - Building a sufficient knowledge of the research domain and, beneficially, understanding the landscape or place of the research in a wider technical ecosystem
- Formulation of focused review questions
- A comprehensive, exhaustive search for primary studies, applying filtering criteria to initial based on search by keywords and conditional search expression
- Quality assessment of included studies
- Identification of the data/information needed to answer the research question
- Data/information extraction – logging in sufficient form to allow back-tracking
- Summary and synthesis of study results (meta-analysis)
- Interpretation of the results to determine their applicability
- Report-writing



Structured SLR for SSE Workflow – As conducted by the PhD student & advised by the advisor (or domain expert)





SLR Workflow/Steps – Brief

Step 0: Preparation and setup

Step 1: Collect papers

- The student searches IEEE, ACM, Scopus, Web of Science, Google Scholar, or other sources.
- KCEG focus: **K1–K2**

Step 2a: Build the candidate set

- The student screens titles, abstracts, and keywords.
- KCEG focus: **K3–K4**

Step 2b: Full-text assessment

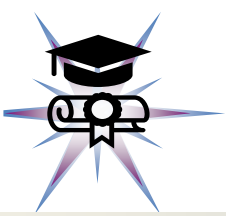
- The student reads the introduction, method, results, discussion, and conclusion.
- KCEG focus: **K4–K8**

Step 3: Build the final SLR set and Quotation/Evidence report/repository

- KCEG focus: **K3–K4, K9**

Field	Student answer
What is the paper about?	
What problem does it address?	
What are the key concepts?	
What exactly is studied?	
What method is used?	
What are the main findings?	
What is one strength?	
What is one weakness?	
How useful is it for my SLR?	
Include as core/support/context/exclude?	

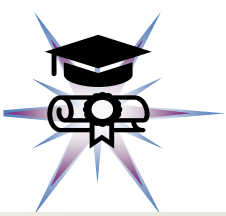
Category		Meaning
Core papers		Directly answer the SLR question
Supporting papers		Help explain concepts, methods, or themes
Context papers		Useful background but not central
Excluded papers		Not relevant or not reliable enough



SLR and Paper assessment and renking – Guiding template

Criterion	Guiding question	Score 0	Score 1	Score 2
Topic relevance	Is this paper about my SLR topic?	No	Partly	Yes, directly
Problem clarity	Can I identify the problem/gap?	No	Vaguely	Clearly
Key concepts	Are the important concepts clear?	No	Somewhat	Yes
Research object	Do I know what is being studied?	No	Partly	Clearly
Method clarity	Do I understand how the study was done?	No	Somewhat	Yes
Evidence quality	Does the paper provide evidence for its claims?	No	Limited	Strong enough
Contribution	Can I say what this paper adds?	No	Vaguely	Clearly
Usefulness for my SLR	Can I use this paper in my review?	No	Background only	Yes, important

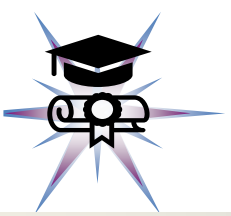
Score	Category	Use
13–16	Core paper	Include in detailed synthesis
9–12	Supporting paper	Use in thematic discussion
5–8	Context paper	Use for background or definitions
0–4	Exclude or keep aside	Not useful for current SLR



SLR Draft – Check points – With view for SLR report

Level	Self-check question	Evidence of progress
1. Search clarity	Did I explain how I found papers?	Databases, search strings, dates, filters
2. Selection clarity	Did I explain why papers were included or excluded?	Inclusion/exclusion criteria
3. Paper understanding	Did I understand what each paper is about?	Terms, concepts, claims, methods
4. Comparison	Did I compare papers rather than only summarize them?	Themes, similarities, differences
5. Quality judgment	Did I judge the strength and limits of the papers?	Method/evidence assessment
6. Synthesis and gap	Did I explain what the literature shows and what is missing?	Gap, future research, contribution

- SLR is not like an educational assignment but a first extended grounding PhD study research
 - To reach the necessary research domain understanding
 - Evidence supported familiarity with the landscape
 - Examples of good papers
- Reader-friendly & reviewer aware
- For internal and future use, create and archive a reference knowledge base
 - Consider returning and re-reading papers – make logs and notes sufficient to find necessary information
 - Possible sharing between team/group members



Structured SLR – Extending SLR *left* and *right*

Preparation & Study/SLR setup: *Extension Left*

- Landscape analysis – Brief summary of the research domain
 - First definition of the Research Questions (e.g., as defined in the PhD study proposal)
- The system model - *To be defined before and re-adjusted during SLR*
 - Entities & Actors/Relations
 - User/Service & Operator (& Auditor)
 - System components: including functionalities, ReqEng, Setup/Config
 - Depending on research domain, can use system design as a starting point, further projected into research, with identifying gaps
 - Structure vs Operation vs Mngnt vs Policy/Monit
 - Lifecycle or Workflow

Performing SLR – Enriched Regular methodology

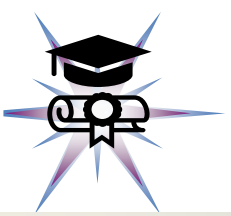
- Paper analysis and ranking
 - Define criteria -> Group criteria by system functions
 - Define assessment levels <- Aligned with KCEG
- Reproducibility as an important assessment criteria for paper
 - Can I reproduce (at what level) and develop further?

ADV – consulted and verified with PhD advisor

Summary/Result/Outcome: *Extension Right (for future after-SSLR work)*

- Report/paper structure
 - Best presented as a dedicated section
 - NOT: I reviewed papers & research domain is presented + some gaps
 - RECOMMENDED (brief for the size of paper):
Defined Landscape + Criteria to support Research QQ (+ADV) + Gaps (ADV) & Identified research direction(s) + Development plan (ADV)
- Reader targeted and reviewer aware

ADV – consulted and verified with PhD advisor



Recommended SSLR structure and stages for doctoral students

Stage A — Preparation and Exploratory scoping

- Purpose: understand the field before claiming to do an SLR.
- 20–40 seed papers,
- initial vocabulary,
- concept map,
- candidate RQs,
- list of venues and databases,
- preliminary taxonomy.

Stage B — Protocol development

- Final RQs,
- search strings (can not only a list of keywords),
- database list,
- inclusion/exclusion criteria,
- quality criteria,
- extraction form

Stage 1 & 2— Systematic search and selection (from records to full-text selection)

- Raw search result table – using database and repositories data
- deduplicated table,

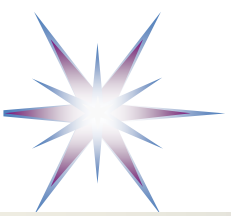
- screening decisions,
- full-text exclusion reasons,
- PRISMA flow diagram,
- final included paper set.

Stage 3 — Extraction and coding (working with full-text)

- Data/information extraction table,
- coding scheme,
- quality assessment table,
- evidence map,
- classification matrix.

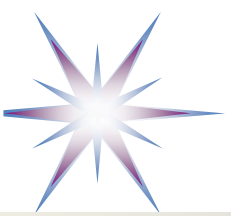
Stage 4 — Synthesis and contribution (summarising value for research)

- Final report and Summary
- taxonomy,
- research gap analysis,
- maturity model or conceptual framework,
- answers to RQs,
- implications for researchers/practitioners,
- future research agenda.



Instructional/methodological support

- Paper review exercise
- Linking to KCEG to papers analysis as a scientific artifact
- Case studies: SLR1 and SLR1



Exercise 1: Detailed Paper Assessment Template (Full)

Paper information

- Paper title
- Publication place and date/year
- What is the paper about in one sentence?
- Main topic / problem addressed:
- Why is this paper relevant to my research?

Basic SLR

2. K1–K3: Vocabulary and factual inventory

- Key terms and phrases
- Main factual claims
- Where are these claims supported?

3. K3–K4: Concepts, entities, and roles

- Main concepts: What are the main concepts used by the authors?
- Definitions or implied meanings: Are the concepts clearly defined? Are any terms used vaguely?
- Entities, objects, or components: What kinds of things exist in the paper's research world?
- Roles and boundaries: What role does each main entity play? What is inside and outside the scope of the paper?

Extended SLR

4. K4-K5: Structure and relations

- Relationship map
- Important relation types

Advanced SLR

5. K6-K8: Methodological reconstruction

- What method does the paper use?
- What are the main steps of the method?
- What data, tools, experiments, models, or analytical procedures are used?
- What is being measured, compared, tested, designed, or evaluated?
- Is the methodology sufficient for the claims made? Why or why not?

6. K7–K9: Inference, assumptions, limitations, and epistemic quality

- Main reasoning chain: How do the authors move from evidence to conclusion? Evidence → interpretation → claim → conclusion:
- Assumptions: What must be true for the paper's argument or method to work?
- Limitations: What are the main limitations, weaknesses, or boundary conditions?
- Uncertainty: What remains uncertain, under-evidenced, or open?
- Evidence quality: Rate the evidence quality:

Advanced but partly used for Basic SLR (as presenting paper results)

7. K9: Contribution and research trajectory

- What is the paper's main contribution?
- What is genuinely new in this paper?
- What is missing or underdeveloped?
- What research question could follow from this paper?
- How could this paper influence my own research trajectory?

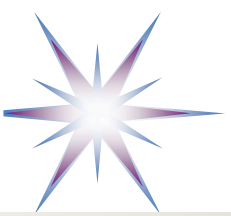


Introductory/Guiding translation of the 9 KCEG levels (orientation)

No need to know philosophical terms such as “ontological framing” or “inferential validity” at the beginning

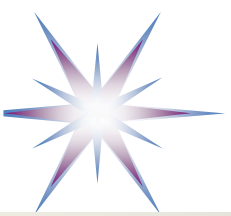
K1-Vocab K2-Syntax K3-Factual/Claims K4-Semantics/Terminology K5-Ontology K6-Structure K7-Inference K8-Method/Model K9-Epistemy/Novelty/Evidence

KCEG level	Definition	Student question	What the student must produce (self-check/check)
K1, K2	Terminological	What are the important words in this paper?	5–10 key terms
K3	Factual / propositional	What does the paper say or claim?	3–5 main claims/findings
K4	Conceptual / semantic	What do the key ideas mean?	Short definitions in the student’s own words
K4	Literature-positioning	Why is this paper written? What problem does it address?	Problem/gap statement
K5	Ontological / object framing	What exactly is being studied?	Object of study: technology, population, method, phenomenon, case, theory
K6	Structural / relational	What is connected to what?	Simple relation map: A affects B, A is compared with B, A is part of B
K7	Inferential / argumentative	What conclusion does the paper reach, and does it follow?	One-sentence argument check
K8	Methodological / validation	How did the authors study it, and can we trust it?	Method and quality note
K9	Contribution / epistemic value	What does this paper add, and what remains missing?	Contribution and limitation note



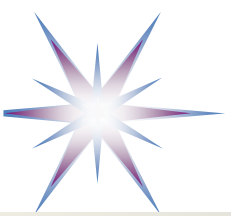
Case Study SLR01 (MS) – Usage control for datasets (AmDex)

- Study type: SLR vs Scoping Review /Mapping Study – SLR more relevant due to historically rich area
- RQs of the PhD study, RQs of SLR
- Necessary diagram – Policy enforcement and usage control for data exchange
 - Includes User/Service & Data Owner (Extended with Data Controller, Data processor and Data Broker)
 - Different from Service access or Process permissions + Data-centric (Access and usage control is linked to data) + Data processing lifecycle + Data Processing workflow
 - Supporting system functionalities/modules: Attributes and properties handling, access and validation; Usage control (linked to services, linked to data)
 - Logging and provenance
- Initial templates for paper analysis and logbook – optimise management, avoid losing information and work
- Initial criteria for papers selection: Databases, venues, journals, authors/organisations, years, citations
- Consider snowballing/provenance due to historically rich area – consider as a source of new solutions to integrate and refactor



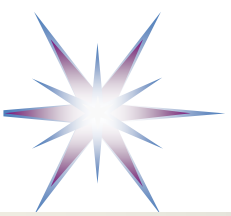
Case Study SLR02 (BA) – ML/LLM Explainability - Accuracy - Energy Efficiency

- Study type: SLR vs Scoping Review /Mapping Study
- RQs of the PhD study, RQs of SLR
- Necessary diagram – process model
 - To represent interaction in Agentic AI, LLM, GenAI: User (context, target, tactic) – Prompt (structure, template) – LLM/Agent answer (request parsing, Model, inference and context/history)
 - LLM/Agentic AI internal structure and supporting infrastructure, data and model management
- Initial templates for papers analysis and logbook – optimise management, avoid losing information and work
- Initial criteria for papers selection: Databases, venues, journals, authors/organisations, years, citations
- Consider snowballing/provenance in limited range – due to a novel and fast developing area

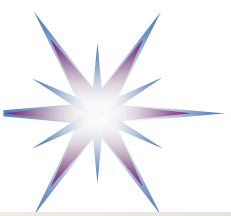


Further development

- Align SLR and Scoping/Mapping Study with KCEG
 - Use as a research skills development
- Provide textual guidelines and templates (Excel, Word, online)
 - Organise the GitHub area to support sharing and contributing discussions
- Recommendations and precautions for using LLM at the initial doctoral study
 - Self-governed and developmental use of LLM/GenAI
 - Recognising the necessary level of domain knowledge for effective use of LLM/GenAI and stimulated research knowledge development



Additional extended slides



Background and Methodological Information

- TODO